SISO Nonminimum Phase System Inversion

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1 Linear Case: The System

Consider a general linear single-input single-output (SISO) system described by the following state equation,

$$\begin{aligned} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) \end{aligned} \tag{1}$$

where the state $x(t)$ has n components.

1.1 Well Defined Relative Degree Assumption

Assumption 1 *Let the system have a well-defined vector relative degree r, i.e.,*

$$\begin{aligned} CA^jB &= 0 \quad if \ j < r-1 \\ CA^{r-1}B &\neq 0 \end{aligned} \tag{2}$$

■

Remark 1 *The relative degree r corresponds to the difference between the order of the denominator and the order of the numerator of the transfer function $G(s)$ of the system (in Eq. 1).*

1.2 Differentiate Output Till Input Appears

The r^{th} derivative of the output is directly related to the input as

$$\begin{aligned} \frac{d^r}{dt^r}y(t) &= CA^rx(t) + CA^{r-1}Bu(t) \\ &= A_yx(t) + B_yu(t) \end{aligned} \tag{3}$$

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1.3 The Inverse Input

From the well-defined relative degree assumption, B_y is invertible (see Eq. 2). We obtain the inverse input from Eq. (3) as

$$u_{inv}(t) = B_y^{-1}\left[\frac{d^r}{dt^r}y(t) - A_y x(t)\right] \tag{4}$$

Remark 2 *We need to know $x(t)$ to compute the inverse input. For minimum-phase systems this can be accomplished using sensors (or state estimators). The poles of the closed loop system are at the zeros of the original system. Therefore, such an approach is not suitable for nonminimum-phase system since it will lead to an unstable closed loop system*

1.4 Decouple State into Known and Unknown Parts

The reduced order inverse to compute the reference state trajectory $x(t) = x_{ref}(t)$ is described, next. The key idea is some components of the state is known when we define the desired output $y_d(\cdot)$ and its time derivatives. In particular, consider the following co-ordinate transformation.

$$\begin{aligned} z(t) &= \begin{bmatrix} y_d(t) \\ y_d^{(1)}(t) \\ \vdots \\ \frac{d^{(r-1)}}{dt^{(r-1)}} y_d(t) \\ \cdots\cdots\cdots\cdots \\ \eta(t) \end{bmatrix} = \begin{bmatrix} \zeta_d(t) \\ \cdots\cdots \\ \eta(t) \end{bmatrix} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{r-1} \\ \cdots\cdots\cdots\cdots \\ T_\eta(t) \end{bmatrix} x(t) = \begin{bmatrix} T_\zeta \\ \cdots\cdots \\ T_\eta \end{bmatrix} x(t) \\ &= Tx(t) \end{aligned} \tag{5}$$

where the bottom portion T_η of the co-ordinate transformation matrix T is chosen such that T is invertible. In the transformed co-ordinates, the known portion of the state is $\zeta_d(\cdot)$ and the unknown portion of the state is η. If a bounded solution to the unknown portion of the state can be found as $\eta_{ref}(\cdot)$, then the co-ordinate transformation Eq. (5) can be used to find the reference state trajectory $x_{ref}(t)$ as

$$
\begin{aligned}
x_{ref}(t) &= T^{-1}\left[\begin{array}{c} \zeta_d(t) \\ \eta_{ref}(t) \end{array}\right] \\
&= \left[T_l^{-1} \,|\, T_r^{-1}\right]\left[\begin{array}{c} \zeta_d(t) \\ \eta_{inv}(t) \end{array}\right] \\
&= T_l^{-1}\zeta_d(t) \;+\; T_r^{-1}\eta_{inv}(t)
\end{aligned}
\tag{6}
$$

1.5 Re-writing the Inverse Input

For simplicity of notation we write the r^{th} time derivative as

$$
y_d^{(r)}(t) = \frac{d^r}{dt^r}y(t).
$$

Then, from Eq. (4) and Eq. (6), the inverse input can be found as

$$
\begin{aligned}
u_{inv}(t) &= B_y^{-1}\left[y_d^{(r)}(t) - A_y x_{ref}(t)\right] \\
&= B_y^{-1}\left[y_d^{(r)}(t) - A_y\left(T_l^{-1}\zeta_d(t) \;+\; T_r^{-1}\eta_{ref}(t)\right)\right] \\
&= B_y^{-1}y_d^{(r)}(t) \;-B_y^{-1}A_y T_l^{-1}\zeta_d(t) \;-B_y^{-1}A_y\; T_r^{-1}\eta_{ref}(t) \\
&= [-B_y^{-1}A_y\; T_r^{-1}]\eta_{ref}(t) \;+\left[(-B_y^{-1}A_yT_l^{-1}) \vdots (B_y^{-1})\right]\left[\begin{array}{c} \zeta_d(t) \\ y_d^{(r)}(t) \end{array}\right] \qquad (7) \\
&= [-B_y^{-1}A_y\; T_r^{-1}]\eta_{ref}(t) \;+\left[(-B_y^{-1}A_yT_l^{-1}) \vdots (B_y^{-1})\right]\mathcal{Y}_d(t) \\
&= [C_{inv}]\eta_{ref}(t) \;+[D_{inv}]\mathcal{Y}_d(t) \qquad (8)
\end{aligned}
$$

1.6 The Internal Dynamics

The unknown component of the state $\eta(t)$ satisfies the following differential equation (found from Eq. 5 by differentiating the equation $\eta(t) = T_\eta x(t)$)

$$
\begin{aligned}
\dot{\eta}(t) &= T_\eta \, \dot{x}(t) \\
&= T_\eta \left[Ax(t) + Bu_{inv}(t) \right] \\
&= T_\eta \left[Ax(t) + B \left(\underbrace{[-B_y^{-1} A_y \; T_r^{-1}] \eta(t) \; + \left[(-B_y^{-1} A_y T_l^{-1}) \vdots (B_y^{-1}) \right] \begin{bmatrix} \zeta_d(t) \\ y_d^{(r)}(t) \end{bmatrix}}_{\text{From Eq.(7)}} \right) \right] \\
&= T_\eta \left[\underbrace{A \left(T_l^{-1} \zeta_d(t) \; + \; T_r^{-1} \eta(t) \right)}_{\text{From Eq.(6)}} + B \left([-B_y^{-1} A_y \; T_r^{-1}] \eta(t) \; + \left[(-B_y^{-1} A_y T_l^{-1}) \vdots (B_y^{-1}) \right] \begin{bmatrix} \zeta_d(t) \\ y_d^{(r)}(t) \end{bmatrix} \right) \right] \\
&= T_\eta \left[(A T_r^{-1}) - B B_y^{-1} A_y \; T_r^{-1}) \right] \eta(t) \quad + T_\eta \left[(A T_l^{-1} - B B_y^{-1} A_y T_l^{-1}) \vdots (B B_y^{-1}) \right] \begin{bmatrix} \zeta_d(t) \\ y_d^{(r)}(t) \end{bmatrix} \\
&= [A_{inv}] \eta(t) \quad + [B_{inv}] \begin{bmatrix} \zeta_d(t) \\ y_d^{(r)}(t) \end{bmatrix} \\
&= [A_{inv}] \eta(t) \quad + [B_{inv}] \mathcal{Y}_d(t)
\end{aligned}
\tag{9}
$$

1.7 The Inverse System

The inverse system is given by (from Eq. 8 and Eq. 9)

$$\begin{aligned} \dot{\eta}_{ref}(t) &= [A_{inv}]\eta_{ref}(t) & +[B_{inv}]\mathcal{Y}_d(t) \\ u_{inv}(t) &= [C_{inv}]\eta_{ref}(t) & +[D_{inv}]\mathcal{Y}_d(t) \end{aligned} \tag{10}$$

which yields both the reference internal state $\eta_{ref}(t)$ and the inverse input $u_{inv}(t)$, and where

$$\begin{aligned} A_{inv} &:= T_\eta[A-(BB_y^{-1}A_y)]T_r^{-1} \\ B_{inv} &:= \left[T_\eta[A-(BB_y^{-1}A_y)]T_l^{-1} \;\vdots\; (T_\eta BB_y^{-1}) \right] \\ C_{inv} &:= -B_y^{-1}A_y\, T_r^{-1} \\ D_{inv} &:= \left[(-B_y^{-1}A_yT_l^{-1}) \vdots (B_y^{-1}) \right] \\ \mathcal{Y}_d(t) &:= \begin{bmatrix} \zeta_d(t) \\ y_d^{(r)}(t) \end{bmatrix}. \end{aligned} \tag{11}$$

Remark 3 *The eigenvalues of A_{inv}, i.e., the poles of the inverse system correspond to the zeros of the original system (Eq. 1). The eigenvalues of A_{inv} are independent of the choice of the matrix T_η (in Eq. 5). Therefore, the inverse system is unstable for nonminimum-phase systems.*

1.8 The Noncausal Inverse for Nonminimum-Phase Case

The internal state η is decoupled into stable and unstable states using a co-ordinate transformation T_{split}

$$T_{split}\begin{bmatrix}\eta_s(t)\\ \eta_u(t)\end{bmatrix} = \eta_{ref}(t) \tag{12}$$

The new state Equations are given by

$$\begin{aligned}\frac{d}{dt}\begin{bmatrix}\eta_s(t)\\ \eta_u(t)\end{bmatrix} &= [T_{split}^{-1}][A_{inv}][T_{split}]\begin{bmatrix}\eta_s(t)\\ \eta_u(t)\end{bmatrix} + [T_{split}^{-1}][B_{inv}]\mathcal{Y}_d(t)\\ &= \begin{bmatrix}A_s & 0\\ 0 & A_u\end{bmatrix}\begin{bmatrix}\eta_s(t)\\ \eta_u(t)\end{bmatrix} + \begin{bmatrix}B_s\\ B_u\end{bmatrix}\mathcal{Y}_d(t)\end{aligned} \tag{13}$$

Remark 4 *The transformation matrix T_{split} is chosen to decouple the internal state η into stable and unstable sub-dynamics. The transformation matrix can be found using the eigenvectors of the matrix $[A_{inv}]$.*

A bounded solution to the decoupled internal state equations

$$\begin{aligned}\dot{\eta}_s(t) &= A_s\eta_s(t) + B_s\mathcal{Y}_d(t)\\ \dot{\eta}_u(t) &= A_u\eta_u(t) + B_u\mathcal{Y}_d(t)\end{aligned}$$

can be found as

$$\begin{aligned}\eta_{s,ref}(t) &= \int_{-\infty}^{t} e^{A_s(t-\tau)}B_s\mathcal{Y}_d(\tau)d\tau\\ &= \text{solved forward in time}\\ \eta_{u,ref}(t) &= \int_{t}^{\infty} e^{-A_u(\tau-t)}B_u\mathcal{Y}_d(\tau)d\tau\\ &= \text{solved backward in time}\end{aligned} \tag{14}$$

The reference internal state trajectory is then found (from Eq. 12 and Eq. 14) as

$$\eta_{ref}(t) = [T_{split}]\begin{bmatrix}\eta_{s,ref}(t)\\ \eta_{u,ref}(t)\end{bmatrix} \tag{15}$$

and the inverse input can be found from Eq. (10) as

$$u_{inv}(t) = [C_{inv}]\eta_{ref}(t) + [D_{inv}]\mathcal{Y}_d(t).$$